	$H = 2.5R$	
3(a)	The rocket experiences equal and opposite gravitational forces exerted by Earth and Moon. Hence there is zero net force as the forces cancel out.	
3(b)	1. Moon does not have an atmosphere hence there is no need to supply energy to do work against air resistance. 2. The gain in gravitational potential energy from Moon to P is much lower than that from Earth to P hence less kinetic energy is required for the rocket to move from Moon to P.	
3(c)	At P, $F_E = F_M$ $\frac{GM_E m}{d_E^{\ 2}} = \frac{GM_M m}{d_M^{\ 2}}$ $M_M = \frac{\left(5.97 \times 10^{24}\right)\left(3.80 \times 10^{7}\right)^2}{\left(3.46 \times 10^{8}\right)^2}$ $= 7.20 \times 10^{22}$ kg	

No.	Solution	Remarks
3(d)	$\phi_P = -\frac{GM_E}{d_E} - \frac{GM_M}{d_M}$ $= -\left(6.67 \times 10^{-11}\right)\left(\frac{5.97 \times 10^{24}}{3.46 \times 10^{8}} + \frac{7.20 \times 10^{22}}{3.80 \times 10^{7}}\right)$ $= -1.277 \times 10^{6}\ \text{J kg}^{-1}$ KE $=$ Loss in GPE $= (7.5)\left[0 - \left(-1.277 \times 10^{16}\right)\right]$ $= 9.58 \times 10^{6}\ \text{J}$	
4(a)(i)	Δx	